

Fabrication, Optimization, and Use of Graphene Field Effect Sensors

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The accurate measurement of chemical concentrations is the foundation for many industrial and medical technologies. While sensitivity, selectivity, and rapid response are all virtues for sensors, the vast range of applications—from managing diabetes to controlling an industrial process to monitoring water quality—precludes a one-size-fits-all solution for chemical or biological samples. Moreover, each strategy for transduction has strengths and weaknesses based on the sensor materials, assay protocols, and sample type that better suit it for either single point measurements or for continuous monitoring. In this review we will discuss the potential of graphene sensors, particularly graphene electronic sensors, to operate as a label-free detection platform that is well suited for real-time measurement of target species. We will discuss how the sensors may be made and the reasons why graphene functions particularly well for these applications. The review has been arranged to discuss each component of the graphene sensor, building from the bottom up. While doing so, we will highlight and review the best practices reported by the many groups active in this field, while pointing out those areas in need of either further development or poised to be an area of major growth.

Since we will discuss in detail using graphene for biosensing, it is useful to briefly discuss the concept of label-free detection. Most detection schemes for biomolecules use a label to enhance the transduced signal. For example, in an enzyme linked immunosorbent assay (ELISA), the target biomolecule is first captured with an antibody, next, an enzyme-labeled antibody binds to a second epitope of the target biomolecule, and finally, the enzyme label turns over a molecular substrate resulting in a visible color change. Because the enzyme can turn over many molecular

substrates, the capture of each target biomolecule results in an amplified signal. Generally speaking, labels can be any material—fluorescent molecules, magnetic particles, quantum dots, etc.—that raises the signature of the biomolecule above the background noise. Use of labels has made possible significant advances in biosensing, many of them commercialized, and can provide rapid, attomolar sensitivity to molecules extracted from fairly complicated matrices.¹ While there are strengths and weaknesses of each of these approaches that the market will ultimately resolve, there are difficulties intrinsic to all labeled assays. The clearest difficulty is the simple fact that a label must be added to the system. Adding the label requires more reagents, often a more complicated fluidics system, and adds complexity to the system by introducing issues of steric hindrance and mass transport. More importantly, these steric hindrances may impact the binding properties of the molecule, altering its natural interaction with the capture probe. Moreover, labels are typically consumed during sensing, which limits either the duration that the sensor may be placed unattended or the frequency with which sensing is performed. These trade-offs are crippling if the goal is real-time monitoring of a compound. An alternative strategy for biosensing is to produce the signal using some aspect of the molecule itself. If this is done, then the signal will be generated when the molecule binds, the signal should be as specific for the target as possible, and there are no consumables to expend. Sensing the molecule itself is a challenging goal but one with a substantial payoff in simplicity, speed, and durability.

Long-term, real-time monitoring of biomolecules would be a significant achievement; however, the benefits will be limited if the cost of the sensor is excessive. Low cost would enable wide dispersal of the sensors for ubiquitous sensing of communicable diseases, for example, or for clinical diagnostics in the developing world. Clearly, cost reduction is a generic goal in sensing, and we know that it requires careful consideration at each level of fabrication—substrate, sensor material, processing, packaging, and electronics. That said, cost is not a scientific goal per se, so we will not address it here beyond noting that chemical vapor deposition (CVD) graphene and graphene oxide (GO) are fairly inexpensive materials amenable to inexpensive processing.

One solution to label-free, real-time sensing are Biomolecular Field Effect Transistors, or BioFETs,^{2,3} which are the intellectual descendants of the ion selective FETs (ISFETs) first reported by Bergveld.⁴ Figure 1 shows that, in conventional field effect transistors, the charge placed on the gate using an electrode

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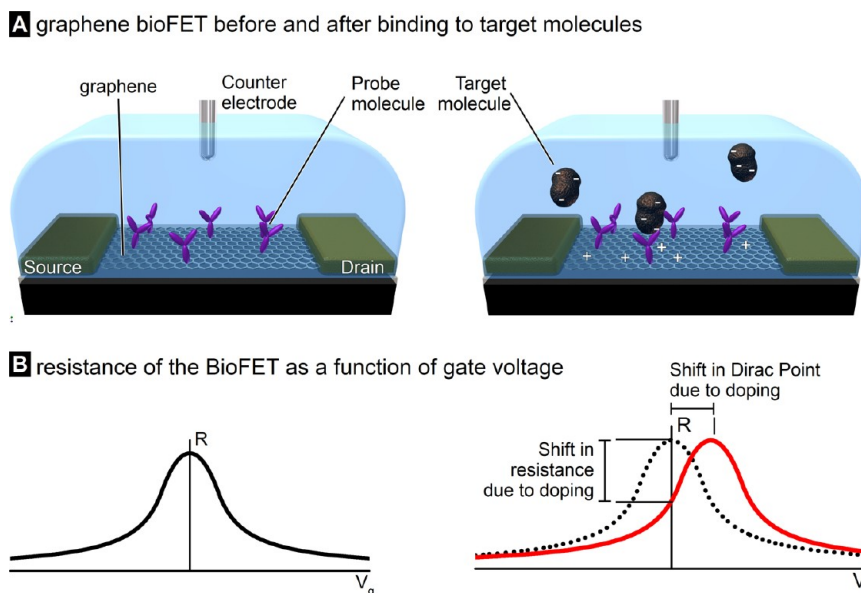


Figure 1. (A) Schematic of a functionalized graphene BioFET before and after binding of the charged target molecule. (B) The binding of the molecule dopes the graphene causing a shift in the Dirac point and consequently a change in the measured resistance.

changes the conductivity of the gate material between the source and drain of the current. Analogously, in a BioFET, it is the intrinsic charge on the gate generated by the binding of a charged molecule or, more generically, charge transfer from an adsorbate that shifts the transconductance of the gate. There are many benefits to this strategy. First, most biomolecules are charged and so the approach is intrinsically label free. Second, the signal is electronic and therefore easily measured, recorded, and reported using conventional electronics. There are no moving parts or optical alignment to worry about, so the devices are often quite rugged. Finally, the manufacturing technology for electronics is exceptionally advanced and thus leads readily both to miniaturization and to scaling up for fabrication.

Multiple groups have developed the FET sensing concept, and the subject has been ably reviewed elsewhere.^{5,6} Particular efforts have been made to develop both Si nanowires and carbon nanotubes (CNTs) for sensing. Both technologies show exceptional performance and are currently under heavy development, so it is difficult at this point to declare definitively any limitations on the technologies. As a general comment, it can be said that obtaining the highest performance from Si nanowires requires many processing steps, which can greatly increase the cost per sensor. Advances in silicon processing should continue to decrease those costs. CNT sensors are more readily obtained; however, one is left with the difficulty either of placing them reproducibly on a substrate or of growing them on a substrate that may have technological limits (inflexible, expensive, etc.). Even when a CNT is placed, its electronic properties depend on very small differences in its diameter and helicity, which are hard to control. Producing mats of semiconducting CNTs is one method demonstrated that circumvents this limitation.⁷

Graphene, however, has many easily accessible strengths. It consists of a single layer of sp^2 bound carbon atoms placed in a honeycomb pattern. It has a high, essentially infinite, surface to volume ratio such that any atom that adsorbs on its surface has the potential to change its electronic properties. This adsorbate can change those properties either by doping the graphene to change the number of carriers or to enhance scattering to reduce its intrinsically high mobility and thus the overall conductivity.

It is compatible with standard semiconductor fabrication technology. Finally, while the first graphene to be studied, exfoliated graphene, was phenomenally expensive due to the requisite extensive user manipulation,⁸ the cost of graphene has since dropped precipitously. CVD can produce very large areas of graphene,⁹ and the cost of chemically modified graphenes such as GO is so low as to make it a nonissue in almost any application. Building a sensor from the bottom up, the first consideration is the substrate. Atomically thin materials such as graphene exhibit a unique property simply due to their thinness—that is, electronic and chemical “semi-transparency.” Early studies of graphene on SiO_2 showed that charged inhomogeneities within the substrate induced local charge variations (or “puddles”) within graphene.¹⁰ Two recent publications^{11,12} have further demonstrated this semi-transparency effect, in which the underlying substrate directs behavior at the top surface of graphene. In one example, the wettability of graphene is strongly influenced by the underlying substrate’s long-range van der Waals forces that extend through the film to impact surface energy.¹² In another example, substrate surface coatings are shown to strongly influence the rate of electron-transfer reactions on the top surface of graphene.¹¹ Multiple solutions have also been proposed to reduce the impact of the substrate. For instance, boron nitride has been shown to be an excellent isolator between graphene and a silicon substrate;¹³ however, the ability to generate high quality, large scale films of the material is still in development. Alternately, one can treat the silicon oxide with octadecyltrichlorosilane¹⁴ to lift the graphene off the silicon oxide and prevent the intercalation of water, which can also dope the graphene. Finally, one could place the graphene on a polymer substrate to avoid entirely these charge inhomogeneities and to gain a flexible substrate.¹⁵ For instance, both PMMA¹⁶ and polystyrene¹⁷ have been shown not to impact the doping of graphene. Together, this indicates that the substrate is a useful knob to tune graphene and that details of the substrate surface must be considered when constructing BioFET devices.

■ GRAPHENE PRODUCTION METHOD

Once the substrate has been prepared, the graphene must be deposited. Rapid developments in graphene synthesis and

growth^{18,19} have resulted in viable routes for the commercialization of graphene-based BioFET sensors and devices. Naturally, for any specific deposition technique, there are benefits and limitations to weigh when considering technology implementation. For BioFET device fabrication, general considerations include substrate choice, film deposition temperature, cleanliness, film adhesion, and film quality, all of which contribute to overall device performance. For graphene materials, synthesis can be broken into two general categories: (i) exfoliation from bulk graphite or (ii) film growth. While many specific recipes exist for each approach, only a subset is well-suited for BioFET applications.

Exfoliation of bulk graphite into individual graphene layers is possible using either mechanical (e.g., physical rubbing) or chemical (e.g., intercalation) forces. Mechanically exfoliated graphene⁸ (Figure 2a) typically exhibits the best electronic and

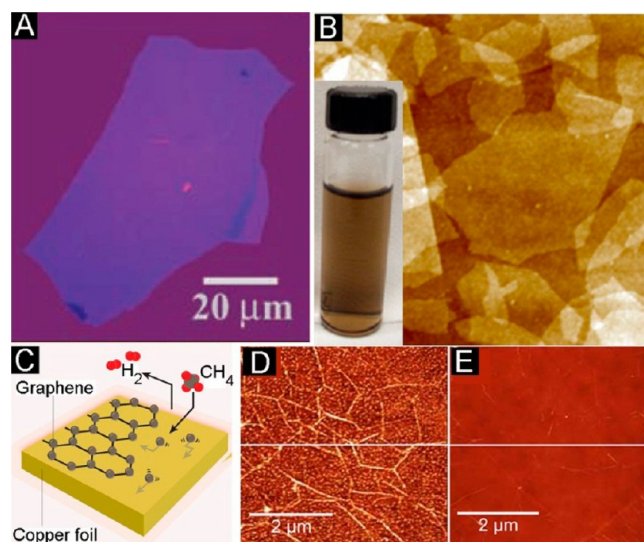


Figure 2. Graphene materials. (A) Optical microscope image of a mechanically exfoliated graphene flake. (Reprinted with permission from ref 8. Copyright AAAS, 2004.) (B) Solution of exfoliated graphene oxide and AFM height image ($2 \times 2 \mu\text{m}^2$) of deposited GO flakes. (C) Schematic illustrating graphene growth on a copper foil. (Reprinted by permission from Macmillan Publishers Ltd: Nature, ref 21, Copyright 2012.) (D, E) AFM height images showing transferred graphene before (D) and after (E) thermal cleaning. (Reprinted with permission from ref 49. Copyright American Institute of Physics, 2011.)

structural quality and, as a result, is most often used to quantify intrinsic properties of the material.²⁰ While the small size and low yield of mechanically exfoliated flakes is not amenable to commercial scaling,²¹ insights into the effects of extrinsic adsorbates on graphene's surface have been explored by using such samples.²² For example, sensing experiments that relate graphene's electronic response to gas adsorption find that charge transfer and scattering are the primary mechanics leading to resistivity changes in the film, similar to that found for carbon nanotubes.²³ While these devices can be exquisitely sensitive to certain adsorbates, they are not selective but rather respond to many electron donating or accepting molecules. Nevertheless, mechanically exfoliated graphene has been used in BioFET construction for experimental devices, showing specific functionality based on biofunctionalization with aptamers²⁴ or antibody fragments.²⁵

GO was the first material used for the production of a graphene-based BioFET by Mohanty and Berry²⁶ and has remained a popular choice among researchers. The advantages of

GO include the low cost of materials, ease of deposition onto various substrates, and the presence of oxygen functional groups, including epoxides and carboxylic acids,²⁷ that can be used for attaching biorecognition molecules.^{26,28,29} Large quantities of GO can be quickly and cheaply produced by chemical exfoliation of graphite via oxidation. This may be achieved using the Hummers method,³⁰ whereby graphite is heavily oxidized with sulfuric acid and potassium permanganate and subsequently exfoliated in an aqueous solution through sonication³¹ or heating.³² This colloidal suspension of GO can be applied to a solid substrate via a number of methods, including spray deposition,³³ vacuum filtration,³⁴ dip coating,³⁵ spin coating (Figure 2b),³⁶ and ink jet printing,³⁷ to form thin, continuous films over large surface areas. Alternately, for BioFETs reported by Kurkina²⁹ and Myung,³⁸ the researchers took advantage of the negatively charged oxygen groups present in GO to selectively pattern graphene films onto positively charged substrate areas.³⁹ Depending on the deposition technique, the thickness of the deposited film can vary; however, there must be at least a few layers of GO flakes (~ 4 nm) to ensure that there are no pinholes in the film. For electronic detection, the insulating GO film must then be reduced to restore conductivity, which can be achieved through heating³⁵ or through chemical reduction with agents such as hydrazine^{33,36,40} or ascorbic acid,³⁷ among others. The use of GO is not without its drawbacks, however, the most notable of which is its high resistance when compared to more pristine forms of graphene.⁴¹ The higher resistance owes to the residual lattice defects created during the oxidation process that can scatter charge carriers and that are unrelated to addition of the biological target to the FET surface. Such scattering can increase the overall noise and may lower the overall sensitivity of the device.

An alternate to GO is graphene synthesis via direct film growth, which produces good material quality with a high degree of scalability. We highlight here graphene grown via chemical vapor deposition (CVD) on transition-metal surfaces,^{18,42–45} as it is relatively inexpensive when compared to materials grown on SiC and can be transferred after growth to arbitrary surfaces. Typical growth recipes include a carbon source such as methane, a growth substrate such as a deposited metal film or a metal foil, and temperatures around 800 to 1000 °C, all of which are held at low (mtorr) or atmospheric pressures.¹⁸ A particularly popular choice for growth substrate is copper because it yields large-area, single-layer graphene due to the low solubility of carbon in copper⁴⁶ (Figure 2c). The resulting graphene films are polycrystalline with grain sizes now approaching 0.1–1 mm in size.^{47,48}

The transfer of graphene from its growth substrate remains an important challenge for advancing BioFET technologies. The most commonly used transfer technique is a wet-chemical process where a polymer, typically PMMA, mechanically stabilizes the graphene film while the metal substrate is etched away^{44,45} using either ammonium persulfate or FeCl_3 . When systematized, this process can transfer very large-areas (>30 in.).⁹ Moreover, the overall process is low temperature (<150 °C), which allows the use of plastic substrates for device fabrication. A drawback to this wet transfer process, however, is carbonaceous residues that can remain even after extensive solvent cleaning (Figure 2d). Such residues impact the physical and electrical properties of the film⁴⁹ (e.g., reducing carrier mobility) and, more critically, can mask subsequent functionalization chemistries. While not as amenable for plastic substrates, a promising route to reduce residues is through annealing at temperatures around 300 to 400 °C^{49–52} (Figure 2e). Recent reports^{53–55} employing CVD graphene in BioFET devices

already show micromolar sensitivities to certain molecules, which indicates that processing residues should not be a major roadblock for advanced sensing. Because of the twin needs for high electronic quality and clean surfaces for biofunctionalization, we expect the development of routine cleaning procedures for graphene to be an area of continued interest for the foreseeable future.

Once the graphene has been placed on the chosen substrate, fabrication of the sensing device often uses standard photolithography techniques. Metal contacts can be deposited directly onto the graphene,²⁴ or the contacts may be formed on the substrate itself with the graphene subsequently deposited on top.²⁸ Patterning of the graphene sheet to form individual, isolated FETs is often accomplished using O₂ plasma.⁵⁶ The conductive electrode surfaces must also be protected from exposure to aqueous solutions to prevent electrolysis and other harmful interactions with the electrodes. Common methods include the use of dielectric materials²⁸ and polymers⁵³ as barrier layers to block interactions with water.

A common issue that must be considered during lithographic patterning is the problem of residual photoresist. Many resists contain aromatic resins, which can strongly adhere to graphene surfaces through π - π stacking.⁵⁷ A sacrificial layer of PMMA may be deposited between the graphene and photoresist to minimize these interactions, although, as discussed previously, PMMA also tends to leave residual contamination. As an alternative to photolithography, FET devices also may be fabricated using inkjet technology.⁵⁸ Indeed, entire graphene sensors have been produced by inkjet, using an ink made from exfoliated GO.³⁷

■ FUNCTIONALIZATION

The next step in biosensor production is to add specificity by attaching the proper biorecognition elements (e.g., antibodies, DNA, peptides). For graphene BioFETs, this has required a considerable amount of research, as graphene is largely inert to many chemical reactions and covalent attachment to the graphene necessarily disrupts its sp² structure, thereby altering its electronic properties. In deference to these issues, some BioFETs have been developed based on noncovalent attachment of the capture probes, thus leaving the electronic structure of the graphene unperturbed. The majority of these, beginning with Ohno²⁴ and Huang,⁵⁴ have taken advantage of physisorbed aromatic molecules that align with the graphene lattice through π - π stacking.⁵⁷ By adsorbing aromatic molecules, such as 1-pyrene butanoic acid, that contain a readily available functional group for subsequent chemical reactions, biomolecular probes can be stably attached to the graphene surface without appreciably disrupting its electronic structure. Another noncovalent method that has been successfully incorporated into BioFETs is the use of adsorbed nanoparticles as anchoring sites for the attachment of probe molecules.⁵⁹ For example, gold nanoparticles are coated onto the graphene surface and subsequently biofunctionalized through thiol chemistry. One of the more novel approaches to noncovalent functionalization has been reported by Cui and colleagues,^{60,61} who used specifically designed peptides that recognize and bind to graphene surfaces (Figure 3). Through the process of phage display, billions of potential binding peptides were screened to find those that show affinity for graphene. The binding of the peptide and graphene, similar to an antibody/antigen interaction, takes place through noncovalent van der Waals interactions, leaving the graphene structure intact. This technique was recently used by Mannoer⁶² to add biological probes to a graphene BioFET for the detection of bacterial cells on tooth enamel. Recently, another noncovalent functionalization

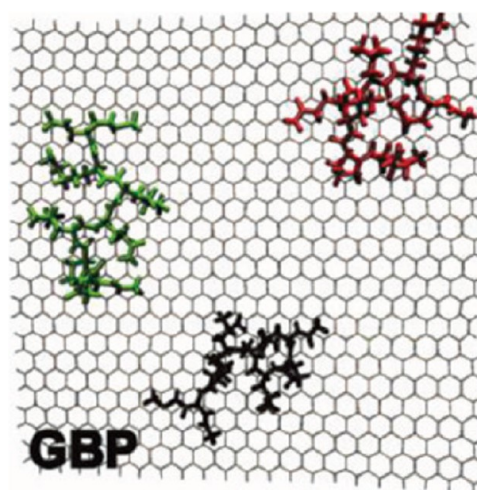


Figure 3. Molecular dynamics simulation showing the lowest energy conformation of a graphene binding peptide. (Reprinted with permission from ref 61. Copyright American Chemical Society, 2011.)

scheme was reported by Park and colleagues,¹⁵ where bilayer graphene is utilized for device production. Here, the top layer of graphene was covalently modified to include specific binding molecules, while the lower graphene sheet was left undisturbed to function as the transduction layer.

While the ability of these approaches to functionalize graphene without disrupting its bonding structure is attractive, they also have several drawbacks. First, adding physisorbed molecules to the surface to serve as an attachment layer for the biomolecular probes moves the biorecognition event farther away from the graphene, leading to greater Debye screening of the target (discussed later in this review) and lower sensitivity.⁶³ Noncovalent attachment also opens up the possibility of desorption of the probe molecules,¹⁵ not only lowering sensitivity but exposing the underlying graphene to possible nonspecific adsorption of biomolecules, leading to false results. As such, methods to covalently attach probe molecules to graphene have also been reported. This was most easily accomplished for sensors using reduced GO as a gate material,^{26,28,29} exploiting the presence of epoxide and carboxyl groups that are already present in the GO flakes as attachment sites. Other methods of covalent graphene functionalization that have been shown to be compatible with BioFET production include exposure to low energy ammonia plasma to form amine groups, which Baraket and colleagues⁵⁵ recently demonstrated for a DNA sensor, and the reaction of graphene with a diazonium salt,^{64–66} which Kasry and colleagues⁶⁷ utilized to attach biotin for the subsequent detection of streptavidin.

A number of additional graphene reactions exist which could potentially prove useful for covalent attachment of biomolecules. Georgakilas⁶⁸ and Quintana⁶⁹ have demonstrated the cyclo-addition reaction of azomethine ylides to graphene. Liu⁷⁰ and Choi⁷¹ reported functionalization of graphene using azide molecules with minimal electronic disruption. Sarkar⁷² reported on the ability of graphene to undergo Diels–Alder reactions, behaving as both a diene or dienophile under various conditions. Huang⁷³ and Yuan⁷⁴ have also recently reported a two-step functionalization procedure, in which graphene is reacted with *n*-butyl lithium, followed by attachment of a brominated target molecule. While none of these methods has of yet been used in a functioning biosensor, they offer several intriguing possibilities for future exploration. Ultimately, any functionalization strategy must strike a balance between the electronic and

chemical properties of graphene to maximize overall sensor performance.

To date, most graphene research has used idealized samples (i.e., just the target and buffer), which allows the change in the FET response to be cleanly explained by target capture. However, as the graphene sensor field matures, issues of nonspecific binding will become increasingly important because real-world samples are much more complex.⁷⁵ For example, it is well-known that biological samples can foul sensor surfaces with an array of proteins, cellular material, and DNA fragments. Consequently, beyond adding specificity, the devices must be chemically modified to prevent nonspecific adhesion. Some researchers have argued that examination of the stochastic signal can help determine the identities of the biomolecules, but the success of these studies has been limited.^{76–78} Significantly, a priori knowledge of the sample composition is needed for effective implementation. In addition to issues of nonspecific binding, switching between a sample and buffer solution can induce changes in pH and ionic concentration above the FET device, both of which are measured as changes.

Only a small number of BioFET papers have progressed far enough into real-world samples to shed significant light on this topic. Kasry and co-workers were using streptavidin-conjugated magnetic beads to bind to their biotinylated FET device and observed no adverse adsorption of the beads to a nonbiotinylated surface.⁶⁷ While informative, it is known that nonspecifically attached, protein-coated magnetic beads can be removed with either magnetic⁷⁹ or fluidic forces,⁸⁰ which lessens the impact of this result. Prevention of nonspecific binding was more definitely observed in Ohno's work.^{24,81} The deposition of aptamer capture agents not only provided specific recognition for immunoglobulin E, but those same negatively charged aptamer strands served as a barrier to nonspecific protein adsorption. Similarly, Mao and co-workers successfully used a blocking buffer with a combination of Tween 20, fish gelatin, and bovine serum albumin (BSA).^{59,82} This classic blocking cocktail coats those sections of surface without capture agents and pacifies them against deleterious adsorption of biomolecules. Nonspecific binding can also be addressed via the wash buffers, with BSA and sodium dodecyl sulfate being the most commonly used constituents.^{26,83,84}

We anticipate that passivation of graphene surfaces will continue to be a key parameter in BioFET device performance. Based on the entirety of the biosensor literature, it is likely a combination of blocking steps and wash steps will be used as researchers gain experience with complex matrices. One promising avenue may be the attachment of polyethylene glycol and other polymers to graphene surfaces.^{85–88} These materials have an established performance history for preventing nonspecific binding in sensor applications and are ripe for incorporation with these FET applications.

■ SENSOR GEOMETRY

One clear advantage of graphene is that, similar to Si or GaAs, it behaves as a thin film with consistent properties over its entire area. This is in contrast to some nanostructures such as single-walled carbon nanotubes (SWCNTs), which can have highly variable properties. There are many benefits to having consistent properties. As our group has published previously, this enables the use of parallel sensors for noise cancellation,²⁸ but there are other benefits that have not been fully explored in the literature. In particular, most sensors to date have been rectangles or irregular flakes of graphene, whereas it is known that long thin sensors generally improve the mass transfer of materials to the surface.⁸⁹ Similarly, molecules in a quiescent solution should

bind first to the perimeter of a round sensor, such that a thin ring shaped sensor should perform quite well compared to conventional disk sensors since the interior regions of the latter would be sparsely filled but still contribute to the baseline of the $\Delta R/R_0$ measurements.⁹⁰ While graphene is readily shaped by a variety of lithographic methods, it is not completely true that the graphene can be shaped arbitrarily without impacting its electronic properties. Indeed, if graphene is shaped into ribbons narrower than ~ 50 nm, then a band gap begins to open in the semi-metal due to localization of carriers across the width of the ribbons.^{91,92} In principle, opening a band gap would benefit sensing, since the shifts in the gate voltage due to biomolecular binding would have a greater effect on the current. To the best of our knowledge, no one has used graphene nanoribbons (GNRs) for biosensing. One can speculate that this is due to the intrinsic difficulty of first forming GNRs with high yields and then having to clear the additional hurdle of forming them reproducibly without leaving resist residue that would block the functionalization. One answer may be the synthetic GNRs reported in 2010 by the Muellen groups where GNRs were grown by polymerizing dibromobisanthryl monomers on a gold surface and then thermally dehydrogenated them to make a nanoribbon three benzene rings wide.⁹³ If such a polymeric approach could be developed outside ultrahigh vacuum (UHV), then consistent electronic performance and side chain biofunctionalization should be quite readily achieved.

A unique aspect of graphene is its bipolar response, or ambipolar transport^{20,94} (e.g., Figure 1b), in which electron or hole conduction dominates depending on the applied gate potential relative to the minimum conduction point, or Dirac point (V_{Dirac}). When extrinsically doped—for example, by a charged molecule— V_{Dirac} shifts to either positive (p-type) or negative (n-type) values (Figure 1b). As such, V_{Dirac} is an exploitable variable to determine the presence of surface charges and has already proven useful in quantifying changes in solution pH.⁹⁵ Simultaneously, graphene's mobility (μ) in the presence of charged impurities (n_{imp}) is reduced by an amount $\mu \propto 1/n_{\text{imp}}$.⁹⁶ Similar to conventional semiconductor materials, the addition of electrodes to probe changes in graphene's conductivity can influence the device output. While graphene is a semi-metal and does not form a traditional Schottky barrier with a metal contact, it can form contact induced states and display significant contact resistance (R_C).^{97–100} The choice of electrode metal is also important because graphene's inert surface can lead to wettability issues or slight work function differences. Typical contacts to graphene include a transition metal wetting layer and Au film (e.g., Ti/Au, 5/30 nm), while specific steps should be taken to reduce R_C .¹⁰¹

■ ELECTRONICS

As we build our sensor from the bottom up, the microfabricated and biofunctionalized device is ready to operate as a field effect transistor, or FET. The transduction mechanism for a FET relies on charge carriers in the channel being influenced by an external, electrostatic perturbation. In a traditional MOSFET device, a gate electrode electrostatically modulates the Fermi energy in the channel material, producing variations in the channel current that are measured through the source-drain electrodes (Figure 4a). In direct analogy, if the solid gate electrode is replaced by an aqueous solution, charged molecules within the medium can modulate conduction in the channel, giving rise to an ion-sensitive FET (ISFET; Figure 4b). First developed in the 1970s,¹⁰² ISFETs with a broad range of channel materials (many with specific surface functionalities; e.g. Figure 4c) have been employed in sensing species including pH,¹⁰³ glucose,^{104,105} streptavidin,¹⁰⁶

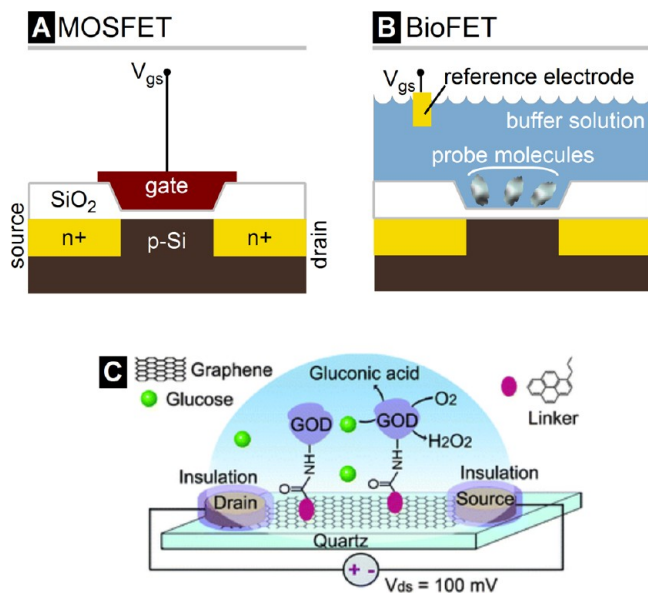


Figure 4. (A) Traditional metal-oxide-semiconductor FET (MOSFET) and (B) Biosensing FET. (C) Scheme illustrating functionalized graphene FET for glucose detection. (Reprinted with permission from ref 54. Copyright The Royal Society of Chemistry, 2010.)

and heavy metal ions,¹⁰⁷ among many others.¹⁰⁸ Multiple mechanisms have been suggested for the perturbation of conduction such as electrostatic gating, changes in gate coupling, carrier

mobility changes, and Schottky barrier effects. In their excellent analysis of these effects, Heller et al. showed that only electrostatic gating and Schottky barrier effects appeared active in carbon nanotube biosensors, although the extent of the effect appeared to depend on the individual device (Figure 5).¹⁰⁹

For conventional MOSFETs operating in the linear region (i.e., such as a resistor), the current (I_D) sensitivity in the channel depends on geometric factors such as the oxide capacitance (C_{OX}), the width (W) and length (L) of the channel, and the mobility of carriers, where

$$I_D = \mu C_{OX} \frac{W}{L} \left[(V_{GS} - V_{th}) V_{DS} - \frac{V_{DS}^2}{2} \right] \quad (1)$$

If we assume validity of the equation, then the advantages of employing graphene as the channel material include the following facts: that μ can be very large (10^3 – 10^4 cm²/(V s)), that graphene is stable in direct contact with aqueous solutions (no gate oxides necessary, so capacitance can be high), and that it is atomically thin and can be shaped into any lateral dimensions. Conventional semiconductors cannot operate without a thin robust oxide protecting the channel, which reduces the gating effect of charged molecules. In addition, while reducing the channel area (e.g., by using carbon nanotubes^{110,111} or nanowires¹¹²) increases the influence of a single adsorption/binding event, this has limiting returns under normal operating conditions.⁹⁰

It is not clear, however, that eq 1 effectively captures the physics of graphene BioFET devices. Indeed, to the best of our knowledge, the ideal electronic properties for a graphene BioFET

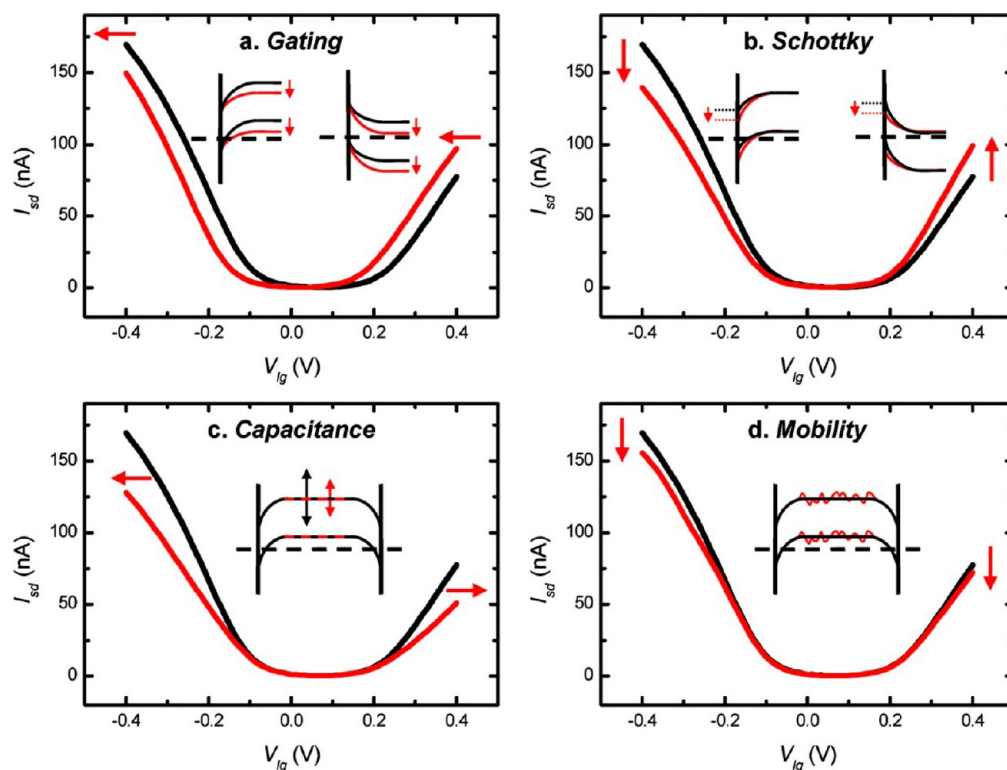


Figure 5. Calculated I – V_g curves before (black) and after (red) protein adsorption on carbon nanotube sensors for four different sensing mechanisms. The bias voltage is 10 mV. (a) Electrostatic gating effect corresponding to a 50 meV shift of the semiconducting bands downward. (b) Schottky barrier effect that corresponds to a change of the difference between metal and SWNT workfunctions of 30 meV. In panels a and b, left and right insets illustrate the corresponding changes in the band diagrams for hole and electron doping respectively. (c) Capacitance mechanism for a 90% coverage of SWNT with protein ($\epsilon = 10$, diameter = 6 nm). In panels c and d, the insets illustrate the corresponding changes in the band diagrams. (d) Mobility mechanism that corresponds to a mobility reduction to a mere 2% of the initial value. (Reprinted with permission from ref 109. Copyright American Chemical Society, 2008.)

have not been derived. Part of the difficulty is that the nature of conduction near the Dirac point (V_{Dirac} , the point of minimum conductivity) remains a matter of debate. However, it has been generally agreed that resistivity away from V_{Dirac} can be reasonably described by

$$\rho(V_g) = \frac{1}{C_g \mu |V_g - V_{\text{Dirac}}|} + \rho_s \quad (2)$$

where C_g is the gate capacitance per unit area, μ is the mobility, V_g is the gate voltage, and ρ_s is the residual resistivity due to screened short-range impurities.¹¹³ Most BioFET measurements track relative resistivity as the indicator of molecular adsorption. Calculating this value, we have

$$\begin{aligned} \frac{\Delta \rho}{\rho} &= \frac{\rho}{\rho_0} - 1 \\ &= \frac{|V_g - V_{D,0}| - |V_g - (V_{D,0} - \Delta V)|}{|V_g - (V_{D,0} - \Delta V)|(1 + C_g \mu \rho_s |V_g - V_{D,0}|)} \end{aligned} \quad (3)$$

where $\Delta V = \Delta Q/C_g$ is the voltage shift in the Dirac point due to the adsorption of the target molecules that bring a charge ΔQ . Some care with this expression needs to be taken, since it is valid neither at $V_g = V_{\text{Dirac}}$ nor at $V_g = V_{\text{Dirac}} - \Delta V$, the positions of the Dirac point before and after doping by the adsorbates. Just off those voltages, however, it is clear that the change in relative resistivity for a given ΔV increases as the carrier mobility and residual resistivity decrease. While a decrease in the residual resistivity is an intuitively obvious improvement, improving the sensitivity by decreasing the mobility runs counter to eq 1 and to assumptions in the community.¹⁵ Sensitivity also generally improves with a drop in the capacitance although the relationship is more complex, again in contradiction to eq 1. A factor that may lessen the expected improvement with lower mobility increases the scattering by short-range scatterers; however, that does not appear to be a strong relation.¹¹³ A similar concern is that a lower mobility may enhance the noise in the system, as discussed below. Ultimately, the graphene BioFET community is still in need of a full treatment of how the unique conductivity of graphene interacts with adsorbates and how to harness that interaction to produce devices that are as sensitive as possible.

A benchmark for biosensors is the minimal detectable level (MDL), which comprises both the sensor response due to an adsorption/binding event and the device noise level. Ideally, the concentration of a charged molecule in solution will be directly proportional to a change in device current (constant voltage mode) or a change in voltage (constant current mode). For single-gated devices (e.g., Figure 4b), the Nernst equation shows a maximum sensitivity of approximately 60 mV/pH units,¹¹⁴ while dual-gated devices in certain regimes can achieve sensitivities as low as approximately 45 mV/pH units.¹¹⁵ These sensitivity limits are further influenced by details of the channel interface, which is most often decorated with specific passivation schemes (or capture probes) to introduce molecular selectivity to a target analyte⁵ (e.g., Figure 4c). Within the biosensor platform there will also be several sources of noise including a biochemical shot noise (Poisson noise) that is inherent in the system and unavoidable, transducer noise,¹¹⁶ and 1/f noise.¹¹⁷ Noise is arguably one of the most important factors to consider in biosensor design, especially since most biosensor tests happen on the order of minutes to hours. Unlike high speed electronic devices, device area is not a primary concern, and as such, the sensing area should be maximized to reduce the impact of 1/f noise. Extrinsic

noise factors can be reduced by introducing a parallel sensor that, for example, experiences identical temperature or solution changes or exposed to the same nonspecific binding events.

■ FLUIDIC SYSTEMS

To apply the BioFET in any practical way, the device must be packaged to suit the intended use. Presently, nearly all reports on graphene BioFETs are essentially research devices using the same basic three electrode, open-reservoir, fluidic measurement systems.^{24,53,118–122} They describe hand-assembled devices that have temporary electrical connections to the source and drain of the graphene film, and a reference electrode within the fluid isolated by a well above the graphene sensor surface (Figure 1A). Clearly, this format does not lend itself to conducting routine assays that require both electronic and fluidic automation with potentially very small (e.g., nanoliter) sample volumes. Investigators are only now addressing the transition of these sensors from the lab to real-world applications that consider speed, cost, and manufacturing constraints.

An important consideration for biological sensing is diffusion-limited detection of low-concentration samples.⁹⁰ Microfluidic systems, with their extremely low dead-volume, fit this need and were implemented in our work, which was one of the first graphene BioFETs designed for DNA detection.²⁸ We constructed a unique microfluidic measurement system that was designed to conveniently allow any number of rGO BioFET sensors to be swapped in and out within seconds (Figure 6).

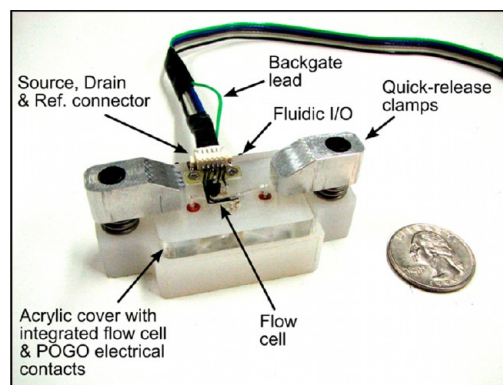


Figure 6. Example of a measurement flow cell designed for rapid switching between BioFET devices.

Both rigid and flexible substrates can be accommodated. The device had two halves that when secured together by spring-loaded clamps would seal two graphene BioFETs (working and reference) in a leak-proof $\sim 2 \mu\text{L}$ flow cell, without adhesives, and make both electrical, with a provision for back gating, and fluidic connections in one step. A platinum counter electrode was exposed to the fluid stream, and the two BioFETs were measured in differential mode to take advantage of the high common mode rejection ratio (CMRR).

In another move toward integration and wearable devices, a more recognizable microfluidic system evoking the lab-on-a-chip platform of the past decade was demonstrated on both glass and PET substrates and capped with a molded polydimethylsiloxane (PDMS) cover with an embedded 800 μm wide channel.¹²³ In this feasibility demonstration, the graphene was not functionalized for biosensing, but it was used to detect flow velocity in the microchannel from 5 to 140 mm/s (on glass). It was also able to show sensitivity to KCl concentration between 1 μM to 10 mM.

Finally, a more sophisticated application was reported in the form of a microcytometer prototype to house a graphene transistor array.¹²⁴ This device captures malaria-infected (*Plasmodium falciparum*) and positively charged red blood cells on endothelial CD36 receptor biofunctionalized CVD grown graphene. Single cells are focused across the transistor array through 8 μm wide SU-8 polymer channels. A PDMS slab with inlet and outlet holes is used to cap the channel. A Ag/AgCl electrode in the flowing liquid top gates the graphene. Their device was able to differentiate between trophozoite and schizonts stages of *P. falciparum* infected cells. Although there are still many basic research challenges that will inform the optimal sensor configuration, these devices are first steps toward commercial devices.

ELECTRICAL DOUBLE LAYER

A primary application for biosensing is medical diagnostics where ideally the biological targets are measured directly in physiological matrices such as blood, serum, plasma, sputum, urine, and stool. The challenge is that these complex matrices not only have a large variety of material components (e.g., proteins, cells, clotting factors) but also contain high salt concentrations. The effect of complex matrices was discussed under the prevention of nonspecific binding; however, the relationship between sensitivity and the ionic concentration of the solution is perhaps the more difficult issue for FET-based sensors.^{53,56} Due to entropy, ions in solution attempt to move apart and so build up at confining surfaces. This leads to a surface charge density, σ , defined by Grahame¹²⁵ as

$$\sigma = \sqrt{8\epsilon\epsilon_0 k_B T c_0} \sinh\left(\frac{zq_e \psi_0}{2k_B T}\right) \quad (4)$$

where ϵ is the dielectric constant of the medium, k_B is the Boltzmann constant, T is the temperature, c_0 is the salt concentration, z is the valency of the salt, and ψ_0 is the surface potential. For values of ψ_0 smaller than ~ 25 meV, eq 4 may be simplified¹²⁶ to $\sigma = (\epsilon\epsilon_0 \psi_0)/(\lambda_D)$ where λ_D , the Debye length, is given by

$$\lambda = \sqrt{\frac{\epsilon\epsilon_0 k_B T}{2q_e^2 c_0 N_A}} \sim \frac{0.3 \text{ nm}}{\sqrt{c_0 \text{ (M)}}} \quad (5)$$

for aqueous solution of monovalent ions. The Debye length is a measure of the electrical double layer formed at the surface, outside of which the charges are heavily screened and so have a minimal impact on the conductance of the FET device (Figure 7). Strictly speaking, this derivation is for a planar sensor; however, as a general rule, the double layer interaction between surfaces or particles of different geometries always decays exponentially with a characteristic decay length equal to the Debye length.¹²⁶ In a physiologically relevant solution, such as 1X PBS, salt concentrations are 150 mM and the resulting Debye length is 0.785 nm.^{25,63} The Debye length for 10-fold and 100-fold reduction in salt only increases the distance to ~ 2.5 and ~ 7.9 nm, respectively.^{25,63} Therefore, only those charges that reside very close to the sensor surface result in a measurable signal, underscoring the necessity of localizing the target molecule as close as possible to the sensor. Importantly, this also means that FET sensors can be indifferent to the bulk solution during sensing and will only respond to the surface docking of biomolecules.

The first solution to the problem of charge screening was to simply exchange the matrix for deionized water after the target

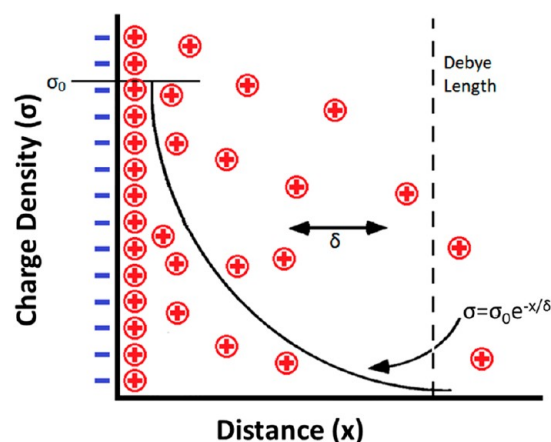


Figure 7. Cartoon of double layer charge accumulation. The negatively charged sensor surface collects a distribution of counterions from the sample solution. The distribution of those charges determines the distance at which the sensor is screened from interaction with additional charged molecules. This distance is also known as the Debye length. (Adapted with permission from ref 127. Copyright, Texas A&M University 2012).

molecule had been captured. However, the flaw in this approach is 2-fold. First, exchange of the buffer solution has major implications for the biological interaction being studied; specifically, a change in stringency from physiological buffer to pure water has been used to discriminate single nucleotide mismatches and to recycle DNA microarrays via a loss in hybridization.¹²⁸ Second, the added steps of buffer exchange add complexity and reagents to the overall use of the sensor, thereby negating some of the advantages gained with the approach.

Other approaches for combating issues of charge screening include the use of ionic liquids to change the device physics. Myung and co-workers used the ionic liquid 1-butyl-3-methylimidazolium hexafluorophosphate to shift the Dirac point of their FET device's top gate and demonstrated the successful detection of cancer markers.³⁸ In another approach, both Mao and co-workers⁵⁹ and Dong and co-workers⁵³ label their bioassays with Au nanoparticles, using the labels to increase the charge at the sensor surface and thereby amplify the signal. In yet a third approach, Stine et al. created FET devices with both a reference sensor and an active sensor functionalized to capture the target molecule. They then expose both sensors to the buffer solutions and take a differential measurement; the differential measurement can subtract out solution effects including pH and ionic charge or even nonspecific binding.^{28,55} Finally, a practical approach to combating the issues of the double layer is to use capture entities that are smaller and therefore bring the target as close as possible to the sensor. Okamoto used F(ab') fragments²⁵ and Ohno used aptamers²⁴ to guarantee that the biointeraction of interest happened close (< 10 nm) to the FET sensor surface.

MEDICAL APPLICATIONS

Now that the various parameters have been considered in the manufacture of a potentially low-cost, reliable, and highly sensitive real-time BioFET sensor, we have finally arrived at their implementation. As noted in the introduction, there are numerous applications where improved biomolecular sensing is desired. For this review, we will focus on medical diagnostics—a natural fit for BioFETs—to highlight the biosensing needs and requirements of various medical specialties. Although by no means exhaustive, the intent of these examples is to introduce

areas in which graphene-based bioFETs can improve medical diagnostics and treatments.

Disease Prevention and Management. It has been a long-standing goal in primary prevention to measure and track metabolic disorders.¹²⁹ Having a point-of-care or in-home capability would enhance screening for preclinical disease.¹³⁰ This will require biosensors with sensitivities capable of detecting preclinical biomarkers and their precursors in a range of matrices including saliva, blood, tissue, urine, and fecal matter. The ability to detect early signs of, for example, elevated prostate-specific antigen (PSA), immunoglobulin E (IgE), or dopamine (DA) levels could positively affect morbidity rates, with the added benefit of lower overall medical costs.

PSA is a protein produced by the prostate gland in men and circulates in the blood. Normal levels of PSA are generally considered to be 4.0 ng/mL, and increasing levels of PSA over time is a sign of prostate cancer. Studies have also shown, however, that prostate cancer can occur both above and below the accepted normal level.¹³¹ A rGO BioFET for prostate specific antigen/ α_1 -antichymotrypsin (PSA-ACT) has been reported.¹¹⁸ This device was built on an aminated glass substrate to which GO mono- to bilayer films were bound by electrostatic interaction. After reduction by hydrazine vapor, the surface was biofunctionalized with PSA-reactive monoclonal antibodies. The functionalized device demonstrated a lower limit of detection (LOD) of 100 fg/mL and a dynamic range of up to 100 ng/mL in 1 μ M buffer solution, and up to 10 ng/mL in human serum.

IgE antibody plays a significant role in the body's immune system, including response to allergens and cancer.¹³² Normal levels are typically between 0.1–0.4 mg/L (0.5–2 nM).¹³³ An aptamer-modified graphene BioFET has been demonstrated to selectively detect human IgE (100 nM) protein from a sample that also contained bovine serum albumin (BSA, 100 nM) and streptavidin (SA, 100 nM).²⁴ Sensitivity for IgE was measured down to 0.29 nM. Special care was taken to account for the Debye length and to avoid adding defects to the single-layer graphene lattice by noncovalently attaching 3 nm aptamer probes (with linker) to the graphene surface via π -stacking. The device is significant for its use of aptamers instead of antibodies, thus eliminating cold-chain requirements.

Dopamine (DA), a catecholamine, is a neurotransmitter produced in the brain. It has a major role in the regulation of reward and movement¹³⁴ and in diseases such as Parkinson's¹³⁵ and schizophrenia.¹³⁶ Normal values for DA in blood are <163 pM (or <30 pg/mL).¹³⁷ A method for large-scale direct patterning of reduced graphene oxide (rGO) on 1.5 cm long substrates, including flexible polyethylene terephthalate (PET) was described.¹²⁰ As a proof of concept, both manually titrated DA (1–60 mM) and direct measurement of vesicular released catecholamines from living neuroendocrine PC12 cells were demonstrated. Although the primary goal of these researchers was to demonstrate a methodology to produce high-yield, low cost rGO FETs using PDMS stamps, the detection results with their use of catecholamines to show device feasibility indicates potential use of BioFETs in neurochemistry measurements.

Finally, although not implemented as a FET, it is worth mentioning the first graphene oxide surface functionalized with a molecularly imprinted polymer (MIP) based on the reversible addition and fragmentation chain transfer (RAFT) polymerization method.¹³⁸ This device demonstrated high affinity for 2,4-dichlorophenol (2,4-DCP), an estrogen surrogate, in the presence of interferents of similar molecular structure. Such a device could have a place for small molecule detection and, similar

to the catecholamine BioFET detector, uses a biorecognition layer that has the potential to eliminate cold-chain requirements. A recently reported GO-MIP hybrid effort operates electrochemically and was able to detect DA with a LOD of 100 nM.¹³⁹ It will be interesting to see whether a graphene–MIP hybrid can be developed into a FET device with faster binding kinetics.

Bacterial Bioburden of Chronic Wounds. There is a desire for rapid detection and identification of pathogenic microorganisms that lead to infections. It has been demonstrated that standard culturing techniques were inadequate at identifying all bacteria in a polymicrobial infection when compared to real-time PCR combined with pyrosequencing.¹⁴⁰ However, PCR methods to identify bacteria can take 2–3 h and will only be successful if the appropriate primers have been developed. Otherwise DNA sequencing becomes necessary for those without primers, and comprehensive identification and quantitation of all microorganisms can require 3–4 days to complete.¹⁴¹ Thus, the choice of using either culture or PCR and sequencing is left to clinicians. An alternative is the quantitative detection of extracellular biomarkers released into the wound by the microorganisms.^{142,143} A cheap, real-time, noncultural solution for early identification of microorganisms can improve clinical outcomes.

The first highly sensitive, but nonspecific, single-bacterium graphene FET has been reported.²⁶ Its operation involved the electrostatic adhesion of a highly negatively charged bacterium (*Bacillus cereus*) to the positively charged aminated graphene oxide sheet resulting in a 42% increase in CMG conductivity. Recently reported is a CVD graphene BioFET that detected in real-time, and with high sensitivity and specificity, various concentrations of *E. coli* as well as the glucose triggered metabolic activities of *E. coli*.¹²¹ The CVD graphene was functionalized with anti-*E. coli* antibodies and demonstrated a sensitivity of 10 cfu/mL and showed insignificant response to negative controls. The BioFET device also distinguished pH changes, proportional to the glucose concentration being fed to *E. coli*, in response to bacterial discharge of metabolic organic acids.

Surgery and Critical Care. Anesthesiologists must negotiate a variety of factors during surgery. Control of stress response¹⁴⁴ and complicated situations such as perioperative glucose management of diabetic patients¹⁴⁵ influence postoperative morbidity and mortality outcomes. As an example, during the application of general anesthesia the blood concentration of pharmaceuticals (e.g., 2,6-diisopropylphenol; Propofol) is highly variable between patients. Real-time blood analysis would lessen the danger of unpredictable dosing and titration variability leading to respiratory arrest.¹⁴⁶ Such real-time analysis could be applicable in the surgical ICU where, in one study, insulin administration to maintain blood glucose levels below 110 mg/dL (6.1 mmol/L) was deemed to reduce morbidity and mortality among both diabetic and nondiabetic critically ill patients.¹⁴⁷ Dysglycemia is a major concern during surgery, and a push for tighter glucose control through frequent monitoring has improperly encouraged the point-of-care use of less accurate devices intended for self-monitoring blood glucose instead of the slower, more expensive central laboratory devices such as blood gas analyzers.¹⁴⁸

One of the first reports of a CVD graphene glucose sensor was constructed emphasizing the use of a flexible polyethylene terephthalate (PET) substrate geared toward wearable or implantable use.¹²² This device had a range from 3.3 to 10.9 mmol/L and is more suited for the potential monitoring of nondiabetic patients.

A simple alternative to the more invasive collection of arterial blood is to analyze breath condensate^{149–151} from intubated patients through an unobtrusive port of the ventilator circuit.

In healthy humans, the concentration of glucose in the nasal and bronchial fluid is <1 mmol/L with breath glucose concentration around 0.40 mmol/L.¹⁴⁹ Glucose or glutamate molecules were detected by a sensor with a CVD graphene film functionalized with glucose oxidase (GOD) or glutamic dehydrogenase (GluD) redox enzymes.⁵⁴ Detection limits were 0.1 mmol/L for glucose and 0.005 mmol/L for glutamate, levels that challenge commonly used electrochemical sensors.

Lactate levels in arterial serum is another effective predictor of outcome in trauma patients in which failure to normalize is associated with a 100% mortality rate.¹⁵² The first flexible lactate CVD graphene sensor has recently been reported.¹¹⁹ CVD graphene was transferred to a PET substrate and was functionalized with lactate oxidase. The device had a linear lactate detection response from 0.08 μ M to 20 μ M. It should be noted that unlike the flexible DA BioFET described earlier,¹²⁰ flexing the substrate did have a noticeable effect on the sensor response due to deformation of the graphene layer.

These examples are just the tip of the iceberg, and as we have just seen, the needs of medical sensing are vast. However, the one unifying theme in all these areas is the desire for sentinel detection capabilities that proactively monitor for biomedically abnormal states.

■ CONCLUSIONS AND FUTURE WORK

Graphene may be the most promising candidate yet for the goal of rapid, sensitive, and inexpensive electronic biosensors. Its unique combination of large area, atomic thickness, and high conductivity offers the possibility of combining exceptional sensitivity to adsorption with large scale mass production. Here, we have covered, in a step by step method, the areas of consideration when designing a BioFET, while highlighting the current and potential future techniques used by researchers in this field. This remains a relatively young research area, and while initial results have shown promise, much work remains to realize a commercially viable diagnostic. One of the difficulties in comparing the techniques used by various researchers is the large number of variables that differ from one study to the next. As a result, no trends are obvious in the literature that point to one particular functionalization chemistry, gate bias, or graphene production method, among other variables, as having a clear advantage. Therefore, one of the most useful areas for future work in the short term may be the systematic study of these areas with stringent controls so that direct comparisons of techniques can be made and best practices established. Another area in need of study is the behavior of the devices exposed to real-world samples, and the subsequent need for surface chemistries to prevent fouling. Be it blood, river water, or cell growth media, complex matrices will undoubtedly challenge BioFETs to maintain specificity in the face of aggressive interferents. While several of the studies discussed herein use basic blocking agents to limit these effects, these may prove insufficient by themselves with real-world samples, particularly where the goal is constant sample monitoring. One final area of exploration is the effect of supporting substrate on the performance of graphene BioFETs. The desire to reduce production cost of these devices will inevitably lead to the search for cheaper substrate materials. However, the transport properties of graphene are strongly affected by interactions at the graphene/substrate interface, and the potential need for back-gating the device should also be considered when choosing a substrate. With ongoing research, graphene BioFETs may prove capable of finally realizing the longstanding goal of sensitive, label-free biodection.

■ AUTHOR INFORMATION

Notes

The authors declare no competing financial interest.

Bioographies

Rory Stine is a contractor with Nova Research, Inc., and is currently contracted for work in the Surface Nanoscience and Sensor Technology Section at the Naval Research Laboratory. Dr. Stine received his B.S. in Chemical Engineering from The Pennsylvania State University in 2001. He went on to obtain his M.S. (2003) and Ph.D. (2005) in Chemical Engineering at Penn State under Dr. Michael Pishko, with research focused on the interaction of biomolecules with surfaces, including the development of nonfouling polymer and biofilm interfaces for biomaterials. He also led a project that focused on developing biosensing devices for potential bioterrorism agents based on cell receptors immobilized within lipid films. While working with Dr. Lloyd Whitman during an American Society of Engineering Education postdoctoral fellowship at the NRL, Dr. Stine worked on novel surface chemistries for biofunctionalizing surfaces in biosensor devices and passivating semiconductor surfaces to improve functional lifetime in infrared electro-optical devices. He worked on the development of biofunctionalization techniques for a wide array of materials, including metal oxides, metal nitrides, and III–V semiconductors, as well as helping to develop a biosensing platform based on GaN/AlGaIn HEMT devices. As a contractor working at NRL, he has worked primarily on the chemical modification of graphene and the production of a biosensing platform using graphene field-effect transistors.

Shawn P. Mulvaney is a Research Chemist for Naval Research Laboratory, Code 6177, the Surface Nanoscience and Sensor Technology Section. Dr. Mulvaney serves as the Assay Team Leader for biosensor development. The multidisciplinary Assay Team develops biocompatible chemistries, assay protocols, and detection strategies while coordinating with engineering for successful integration of assays with system hardware. Dr. Mulvaney has published >20 papers in the field of biosensors, including being recognized by Elsevier for offering the best paper presented at the 10th World Congress at Biosensors. Dr. Mulvaney received an ACS certified B.S. in Chemistry in 1997 from The College of William and Mary. Graduate work in vibrational spectroscopy and the creation of nanoscale architectures culminated in a Ph.D. in 2001 from The Pennsylvania State University. His current interests in biosensor development were cultivated during a NRC Postdoctoral Fellowship at NRL.

Jeremy T. Robinson earned his Bachelor's degree in Physics from Towson University in 2002 and went on to study Materials Science and Engineering at UC Berkeley. Upon finishing his Ph.D. in 2007 he worked for one year as an NRC postdoctoral fellow in the Electronics Science Division at the Naval Research Laboratory in Washington, DC. He joined the full-time staff at NRL in 2008, where he currently resides. His current research activities focus on the development of graphene-based materials including studying graphene's electronic, mechanical, and chemical properties.

Cy R. Tamanaha studied Biomedical Engineering at Worcester Polytechnic Institute (U.S.A.), where he received a M.E. and a Ph.D. in 1994 and 1997, respectively. He completed an ASEE Postdoctoral Fellowship at the U.S. Naval Research Laboratory in 2000 and then worked as a Senior Scientist with Geo-Centers, Inc. until 2003. Since 2003, he has been a staff Research Engineer in the Surface Nanoscience and Sensor Technology Section (Code 6177) of the U.S. Naval Research Laboratory. As Sensor Technology Project Leader for Code 6177, he is the point of contact for the group's optical, magneto-electronic, and chem/bio sensor R&D and commercialization efforts. His current research activities include developing advanced graphene-based BioFET sensors for near real-time, label-free detection and identification of biological threats and biomarkers.

Paul E. Sheehan currently heads the Surface Nanoscience and Sensor Technology Section at the U.S. Naval Research Laboratory. Dr. Sheehan was a University Fellow at the University of North Carolina where he received a B.S. in Chemistry-Based Materials Science in 1993. He earned a Ph.D. in Chemical Physics with Professor Charles Lieber at Harvard University in 1997 studying the nanotribology of solid lubricants and the mechanics of inorganic nanowires and carbon nanotubes. In 2005, this latter work was named one of the top 10 papers in Materials Science for the past decade by ISI. During a National Research Council fellowship with Dr. Rich Colton at the Naval Research Laboratory, he helped develop a DNA sensor based on magnetoelectronics that remains one of the most sensitive and robust means available of detecting biomolecules. His interests in nanoscience are far-ranging, and he has published on chemical templates for assembling nanoscale objects, the impact of diffusion on nanoscale biosensors, why magnetotactic bacteria swim north, and directly writing graphene circuitry with hot AFM probes. His current interest is using chemically modified graphene as a universal strategy for surface modification.

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